

Forcing beats fitting: the minimal generative-evidence case for BST (one page)

Why the same five integers that name one geometry force the
Standard Model's dimensionless content — and how to tell forcing
from fitting in one question

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Forcing beats fitting — the one-page case

The distinction that decides everything

A theory that *reproduces* N physical constants may still have *fitted* them. The only question that separates derivation from numerology is where the physics enters: upstream (in the premise, fixed before you look at the data) or downstream (in the output, tuned to match). Wyler (1969) reproduced α to high precision and is remembered as numerology — not because the number was wrong, but because his construction anchored on the target. BST's claim is that its anchor is target-innocent: the premise is one geometry, fixed by its own mathematics, and the constants fall out downstream.

The premise (fixed target-innocently, before any observable)

One bounded symmetric domain, $D_{IV}^5 = SO_0(5,2)/[SO(5)\times SO(2)]$, selected by internal criteria (it is the unique automorphism-invariant Born space of the required type). Its invariants are five integers — $N_c = 3$, $n_C = 5$, $C_2 = 6$, $g = 7$, $N_{\max} = 137$ (with rank = 2) — each a property of the geometry (root multiplicities, dimensions, Casimir), none read from a physical measurement. That is the whole input. There are zero dimensionless parameters tuned to the data, and — a theorem, not a shortfall — exactly one dimensionful input (the electron mass; a dimensionless set cannot produce a quantity with units, so one scale is the floor every theory pays, GR with G , the SM with v). We name the one open question at the premise itself: the domain's own dimension, $n_C = 5$. The selection criteria narrow the domain sharply, but whether $n_C = 5$ is fully *forced* or enters as a primitive label — the honest referee's “where is this five in the definition?” — is the frontier of the premise,

and we mark it rather than bury it. (The five *derived* integers are properties of the domain once chosen; this question is one level up, about the choice.)

The minimal generative evidence (a few sharp, target-innocent, over-determined)

observable	forced form (geometry only)	value	status
m_p/m_e	$6\pi^5 =$	1836.12 vs	Derived
α^{-1}	$N_c! \cdot \pi^{n_C}$	1836.15 (0.002%)	Identified-strong
	$N_{\max} =$	137 (+0.036	
	$N_c^3 \cdot n_C + \text{rank}$	curvature)	
Newton's G	reduces to $m_e +$ α (2 inputs $\rightarrow$ 1)	0.065%	Derived-structure (one relation, several readings — counted once)

The G reduction rides on α through 24 powers of it ($m_{\text{Planck}}/m_e = 6\pi^5 \cdot \alpha^{12}$; $G \propto m_{\text{Planck}}^{-2}$). That α is the *Identified* α of the row above — measured atomically (recoil, $g-2$), G-independent — not derived: the reduction is real and G-independent, but the coupling it rides on is a measured number, and we tier it as such.

These are not three lucky hits. The same five integers force these and ~150 more, across independent sectors (nuclear, electroweak, PMNS/CKM, cosmology). When one integer fixes many observables, that is a single constraint with many readings — a web, not N independent fits — and it is the opposite of what a fitted model can do (a fit spends one parameter per number). We state this up front, per our own discipline, so a “new” prediction is checked as *another reading of the same object*, not a fresh win — and, just as strictly, we flag where the web is *not* yet one object. (Example, at its honest limit: for the charged leptons the three mass *numbers* satisfy one exact relation — Koide's $Q = 2/3$, to 0.001% — but that relation is currently *identified*, not yet derived, and the individual mass *forms* are separate identifications, not readings of a single equation. The muon's $(24/\pi^2)^6$ is the one open form; deriving it and the Koide filling-amplitude would close the charged-lepton sector on the electron mass alone. We do not double-count the tau's 49.71 form — a mass the mass-chain already yields. A patchwork we call a patchwork.)

The one-question test a reviewer can apply

For each claimed derivation: does its *derivation* reference the target value, or only the geometry? If only the geometry (target-innocent integers, no look-elsewhere), it is forced. If the target appears in the anchoring, it is fitted, and we label it so. Every load-bearing claim in BST carries this provenance tag (Derived / Identified / Conditional / Structural).

The instrument is trustworthy because it cut its own count

The discipline that produces these labels is falsifiable *in use*: applying it lowered our own honest “derived” count ($21 \rightarrow 12$) while the accuracy of the numbers never moved. A device that reduces its makers’ claims and keeps the data fixed is a working instrument, not an advocacy tool — most of the rules were written from a case where the rubric caught its own authors mid-over-claim. We publish our open ledger (every soft spot at tier) and our honest boundaries — and behind the sharpest one is now a theorem. Decomposing the finite internal space of the geometry under its symmetry $K = SO(5) \times SO(2)$ is exact: it splits into three pieces landing on the three real division algebras (Frobenius: $\mathbb{R}, \mathbb{C}, \mathbb{H}$ are the only ones) — a complex charge circle, a quaternionic spinor, and a real three-dimensional color space. This single reality-type census *explains* the Standard Model’s gauge structure. A complex or quaternionic space already carries a complex structure (an “i”; a quaternionic J has $J^2 = -1$), so its unitary symmetry is intrinsic — $U(1)$ and $SU(2)$ are DERIVED, read off the geometry. A real space carries no such structure: the color three-space gives only a *geometric* $SO(3)$ rotation of the three color directions (a spacetime-like symmetry, not an internal gauge) plus a discrete $\mathbb{Z}_2$ — so the internal gauge algebra is exactly $U(1) \times SU(2) \times \mathbb{Z}_2$, with no internal color at all, and the whole of $SU(3)$ (the eight gluons) is imported. That is the honest, sharper boundary: the geometry *derives electroweak* and *explains why color cannot be geometrized the same way* (a real space has no complex structure to gauge), while $SU(3)$ is entirely the gauge principle — the standard input every derivation of the Standard Model takes. The same three-color object carries the integer $N_c = 3$ that appears in the weak mixing angle ($\sin^2 \theta_W \approx 3/13$, $\sim 0.2\%$; a loose tie of a shared object, not a forced angle) and baryon number (a topological winding). The credential is now theorem-backed: BST does not *assume* the Connes finite algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ — it reads it off one geometry, and in doing so tells you why two gauge factors are forced and one imported. We claim exactly that — electroweak derived, color imported, with the reason — and never “the geometry forces the dynamics.”

The exposure (generative theories forbid; fits don’t)

BST makes forbidding predictions — the Five-Absence set: no supersymmetry, no proton decay, no dark-matter particle, no magnetic monopoles, no sterile neutrinos, no grand unification. Each follows from the irreducibility of the one geometry; any single positive detection falsifies the framework. A fitted model can always absorb a new particle; a forced one cannot. That asymmetry is the point.

Companion to Paper A (the derivations) and Paper B (the full 17-rule Forcing+Evidence standard). This one page is the minimal case: premise fixed target-innocently, a handful of sharp forced numbers, the over-determination web, the one-question test, the count-cutting instrument, and the forbidding falsifiers. Reproduction is not derivation; the difference is upstream vs downstream, and it is checkable in one question.